What Is

Agile? is an iterative approach to project management and software development that helps teams provide value to their customers faster by delivering work in small, but consumable, increments. Requirements, plans, and results are evaluated continuously so teams have a natural mechanism for responding to change quickly.

Scrum? is one of the most popular frameworks for implementing agile. With scrum, the product is built in a series of fixed-length iterations called sprints that give teams a framework for shipping on a regular cadence.

Agile Software Development Manifesto

Individuals and interactions over processes and tools

Working software over comprehensive documentation

Customer collaboration over contract negotiation

Responding to change over following a plan

4 Agile Values : https://www.atlassian.com/agile/manifesto

Principles Behind The Agile Manifesto

Our highest priority is to satisfy the customer through early and continuous delivery of valuable software.

Welcome changing requirements, even late in development. Agile processes harness change for the customer's competitive advantage.

Deliver working software frequently, from a couple of weeks to a couple of months, with a preference to the shorter timescale.

12 Agile Principles : https://agilemanifesto.org/principles.html

Principles Behind The Agile Manifesto

Business people and developers must work together daily throughout the project.

Build projects around motivated individuals. Give them the environment and support they need, and trust them to get the job done.

The most efficient and effective method of conveying information to and within a development team is face-to-face conversation.

Principles Behind The Agile Manifesto

Working software is the primary measure of progress.

Agile processes promote sustainable development. The sponsors, developers, and users should be able to maintain a constant pace indefinitely.

Continuous attention to technical excellence and good design enhances agility.

Principles Behind The Agile Manifesto

Simplicity--the art of maximizing the amount of work not done--is essential.

The best architectures, requirements, and designs emerge from self-organizing teams.

At regular intervals, the team reflects on how to become more effective, then tunes and adjusts its behavior accordingly.

Scrum Theory

Scrum is founded on empiricism and lean thinking. Empiricism asserts that knowledge comes from experience and making decisions based on what is observed. Lean thinking reduces waste and focuses on the essentials.

Scrum employs an iterative, incremental approach to optimize predictability and to control risk. Scrum engages groups of people who collectively have all the skills and expertise to do the work and share or acquire such skills as needed.

Scrum combines four formal events for inspection and adaptation within a containing event, the Sprint. These events work because they implement the empirical Scrum pillars of **transparency, inspection, and adaptation**.

Transparency

The emergent process and work must be visible to those performing the work as well as those receiving the work. With Scrum, important decisions are based on the perceived state of its three formal artifacts.

Artifacts that have low transparency can lead to decisions that diminish value and increase risk.

Transparency enables inspection. Inspection without transparency is misleading and wasteful.

Inspection

The Scrum artifacts and the progress toward agreed goals must be inspected frequently and diligently to detect potentially undesirable variances or problems.

To help with inspection, Scrum provides cadence in the form of its five events.

Inspection enables adaptation. Inspection without adaptation is considered pointless. Scrum events are designed to provoke change.

Adaptation

If any aspects of a process deviate outside acceptable limits or if the resulting product is unacceptable, the process being applied or the materials being produced must be adjusted. The adjustment must be made as soon as possible to minimize further deviation.

Adaptation becomes more difficult when the people involved are not empowered or self-managing.

A Scrum Team is expected to adapt the moment it learns anything new through inspection.

Scrum Values

Successful use of Scrum depends on people becoming more proficient in living five values: Commitment, Focus, Openness, Respect, and Courage (C-FORC).

The Scrum Team **commits** to achieving its goals and to supporting each other.

Their primary **focus** is on the work of the Sprint to make the best possible progress toward these goals.

The Scrum Team and its stakeholders are **open** about the work and the challenges.

Scrum Values

Scrum Team members **respect** each other to be capable, independent people, and are respected as such by the people with whom they work.

The Scrum Team members have the **courage** to do the right thing, to work on tough problems.

The Scrum Team members learn and explore the values as they work with the Scrum events and artifacts.

Scrum Team

The fundamental unit of Scrum is a small team of people, a Scrum Team.

The Scrum Team consists of one Scrum Master, one Product Owner, and Developers.

Within a Scrum Team, there are no sub-teams or hierarchies.

It is a cohesive unit of professionals focused on one objective at a time, the Product Goal.

Scrum Team

Scrum Teams are cross-functional, meaning the members have all the skills necessary to create value each Sprint.

They are also self-managing, meaning they internally decide who does what, when, and how.

The Scrum Team is small enough to remain nimble and large enough to complete significant work within a Sprint, typically 10 or fewer people.

Scrum Team

The Scrum Team is responsible for all product-related activities from stakeholder collaboration, verification, maintenance, operation, experimentation, research and development, and anything else that might be required.

They are structured and empowered by the organization to manage their own work. Working in Sprints at a sustainable pace improves the Scrum Team's focus and consistency.

The entire Scrum Team is accountable for creating a valuable, useful Increment every Sprint. Scrum defines three specific accountabilities within the Scrum Team: the Developers, the Product Owner, and the Scrum Master.

Developers

The Developers are always accountable for:

- Creating a plan for the Sprint, the Sprint Backlog.
- Instilling quality by adhering to a Definition of Done.
- Adapting their plan each day toward the Sprint Goal. and,
- Holding each other accountable as professionals.

Product Owner

The Product Owner is accountable for maximizing the value of the product (ROI).

The Product Owner is also accountable for effective Product Backlog management, which includes:

- Developing and explicitly communicating the Product Goal.
- Creating and clearly communicating Product Backlog items.
- Ordering Product Backlog items. and,
- Ensuring that the Product Backlog is transparent, visible and understood.

Scrum Master

The Scrum Master serves the Scrum Team in several ways, including:

- Coaching the team members in self-management and cross-functionality;
- Helping the Scrum Team focus on creating high-value Increments that meet the Definition of Done;
- Causing the removal of impediments to the Scrum Team’s progress; and,
- Ensuring that all Scrum events take place and are positive, productive, and kept within the timebox.

Scrum Master

The Scrum Master serves the Product Owner in several ways, including:

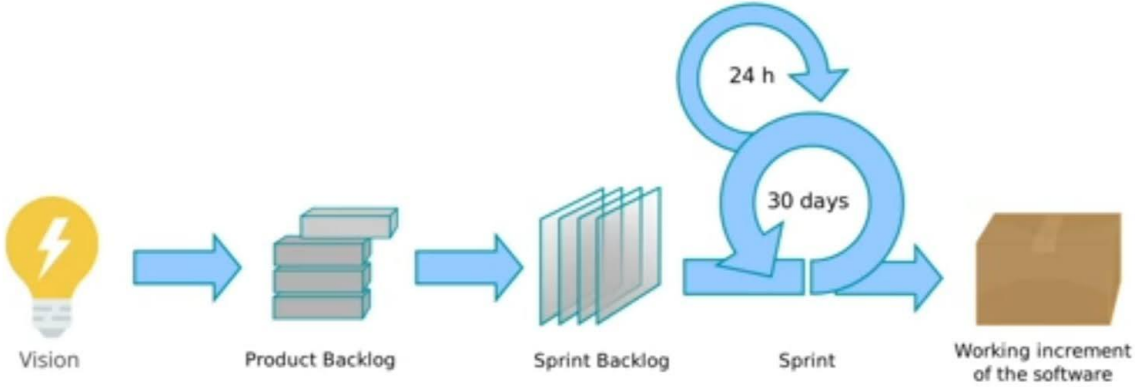
- Helping find techniques for effective Product Goal definition and Product Backlog management;
- Helping the Scrum Team understand the need for clear and concise Product Backlog items;
- Helping establish empirical product planning for a complex environment; and,
- Facilitating stakeholder collaboration as requested or needed.

Scrum Master

The Scrum Master serves the organization in several ways, including:

- Leading, training, and coaching the organization in its Scrum adoption;
- Planning and advising Scrum implementations within the organization;
- Helping employees and stakeholders understand and enact an empirical approach for complex work;
- Removing barriers between stakeholders and Scrum Teams.

Scrum Overview



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Scrum Projects

Accomplished through:

- Themes and Epics
- Releases
- Sprints
- User Stories

Scrum Events : Specific Purpose

Sprint: Container event (30 days or less in duration).

Sprint Planning : From - Product backlog. To - Sprint goal, Sprint backlog.

Daily Scrum : From - Daily progress. To - Updated daily plan.

Sprint Review : From - Sprint, Increment. To - Updated product backlog.

Sprint Retrospective : From - Past sprint. To - Improvements to next Sprint.

The Sprint

The Sprint is a container for all other events.

Sprints are the heartbeat of Scrum, where ideas are turned into value.

They are fixed length events of one month or less to create consistency.

A new Sprint starts immediately after the conclusion of the previous Sprint.

All the work necessary to achieve the Product Goal, including Sprint Planning, Daily Scrums, Sprint Review, and Sprint Retrospective, happen within Sprints.

A Sprint could be cancelled if the Sprint Goal becomes obsolete. Only the Product Owner has the authority to cancel the Sprint.

Sprint Planning Meeting

Attendees Required: development team, scrum master, product owner.

When: At the beginning of a sprint.

Duration: Usually two hours per week of iteration – e.g. a two-week sprint kicks off with a four-hour planning meeting. The meeting ends when its purpose has been achieved.

Purpose: Plan the work of the sprint. The team agrees to the sprint goal and the sprint backlog (prioritized product backlog).

* The Scrum Team may also invite other people to attend Sprint Planning to provide advice.

Sprint Planning Meeting

Why is this Sprint valuable?

The Product Owner proposes how the product could increase its value and utility in the current Sprint.

The whole Scrum Team then collaborates to define a Sprint Goal that communicates why the Sprint is valuable to stakeholders.

The Sprint Goal must be finalized prior to the end of Sprint Planning.

Sprint Planning Meeting

What can be Done this Sprint?

Through discussion with the Product Owner, the Developers select items from the Product Backlog to include in the current Sprint.

The Scrum Team may refine these items during this process, which increases understanding and confidence.

Selecting how much can be completed within a Sprint may be challenging. However, the more the Developers know about their past performance, their upcoming capacity, and their Definition of Done, the more confident they will be in their Sprint forecasts.

Sprint Planning Meeting

How will the chosen work get done?

For each selected Product Backlog item, the Developers plan the work necessary to create an Increment that meets the Definition of Done.

This is often done by decomposing Product Backlog items into smaller work items of one day or less.

How this is done is at the sole discretion of the Developers. No one else tells them how to turn Product Backlog items into Increments of value.

Sprint Planning Meeting
Inputs from Stakeholders
Product Owner
Features
Product Backlog
Selects starting from the top as much as it can commit for the Sprint
Sprint Planning Meeting
Development Team
Scrum Master
Burn-Downs/Burn-Ups
Daily Scrums
Every 24 Hours
1-4 Week Sprint
Task Breakout
Sprint Backlog
Items selected during the Sprint and the Sprint end date do not change
Finished Work

Sprint Planning Meeting

- Product Backlog
- Team Capacity based on Past Performance
- Business Conditions
- Technology Stability
- Most Recent Product Increment
- Current Status of the Project

→

1. Review
2. Consider
3. Organize

→

1. Sprint Goal
2. Sprint Backlog
3. Sprint Plan

Daily Scrum (Standup) Meeting

Attendees: (Primarily) development team

When: Once per day, typically in the morning

Duration: No more than 15 minutes. Don't book a conference room and conduct the stand-up sitting down. Standing up helps keep the meeting short!

Purpose: The daily standup is designed to inform everyone quickly of what's going on across the team and to plan the work of the day. It's not a full status meeting. The tone should be light and fun, but informative. Have each team member answer the following questions:

- What did I complete yesterday?
- What will I work on today?
- Am I blocked by anything?

Daily Scrum (Standup) Meeting

There's an implicit accountability in reporting what work you completed yesterday in front of your peers. No one wants to be the team member who is constantly doing the same thing and not making progress.

If the Product Owner or Scrum Master are actively working on items in the Sprint Backlog, they participate as Developers.

The Daily Scrum is not the only time Developers are allowed to adjust their plan. They often meet throughout the day for more detailed discussions about adapting or re-planning the rest of the Sprint’s work.

ProTip: You can use the active sprints of your scrum board during the daily standup, so that each member can view the tasks they're working on.

Sprint Review Meeting

Attendees: (Primarily) development team, scrum master, product owner. (Optional) stakeholders.

When: Typically on the last day of the sprint.

Duration: Typically two hours for a two-week sprint.

Purpose: Inspect the increment and collaboratively update the product backlog.

Questions to ask:

Did the team meet the sprint forecast?

Was there work added or removed during the middle of the sprint?

Did any work not get completed within the sprint? If so, why?

Sprint Review Meeting

The Scrum Team presents the results of their work to key stakeholders and progress toward the Product Goal is discussed.

During the event, the Scrum Team and stakeholders review what was accomplished in the Sprint and what has changed in their environment. Based on this information, attendees collaborate on what to do next.

The Product Backlog may also be adjusted to meet new opportunities.

The Sprint Review is a working session and the Scrum Team should avoid limiting it to a presentation.

Sprint Retrospective Meeting

Attendees: development team, scrum master, product owner.

When: At the end of an iteration.

Duration: Typically 90 minutes for a two-week sprint.

Purpose: The team inspects itself, including its processes, tools and team interaction. Improvement issues are often added to the next sprint's backlog.

Questions to ask:

- What did we do well during the sprint?
- What could we have done better?
- What are we going to do better for next time?

Sprint Retrospective Meeting

Retrospectives aren't just a time for complaints without action. Use retrospectives to find out what's working so the team can continue to focus on those areas. Also, find out what's not working and use the time to find creative solutions and develop an action plan. Continuous improvement is what sustains and drives development within an agile team, and retrospectives are a key part of that.

After you complete the sprint, have your team do a retrospective. Document your retrospective somewhere.

At the end of the sprint, you must complete it. If the sprint has incomplete issues, you can:

Move the issue(s) to the backlog.

Move the issue(s) to a future sprint.

Move the issue(s) to a new sprint.

ProTip: Even if things are going well across the team, don't stop doing retrospectives. Retrospectives provide ongoing guidance for the team to keep things going well.

Sprint Retrospective Meeting

The purpose of the Sprint Retrospective is to plan ways to increase quality and effectiveness.

The Scrum Team inspects how the last Sprint went with regards to individuals, interactions, processes, tools, and their Definition of Done.

Inspected elements often vary with the domain of work.

Assumptions that led them astray are identified and their origins explored.

The Scrum Team discusses what went well during the Sprint, what problems it encountered, and how those problems were (or were not) solved.

The Scrum Team identifies the most helpful changes to improve its effectiveness.

The most impactful improvements are addressed as soon as possible. They may even be added to the Sprint Backlog for the next Sprint.

The Sprint Retrospective concludes the Sprint. It is timeboxed to a maximum of three hours for a one month Sprint. For shorter Sprints, the event is usually shorter.

Time-boxing

Maximum Duration:	
Sprint	2 to 4 weeks
Sprint Planning	2 hours for each week of the Sprint
Daily Scrums	15 minutes
Sprint Reviews	1 hour for each week of the Sprint
Sprint Retrospective	45 minutes for each week of the Sprint

Product Backlog

Commitment: Product Goal

The Product Goal describes a future state of the product which can serve as a target for the Scrum Team to plan against.

The Product Goal is in the Product Backlog.

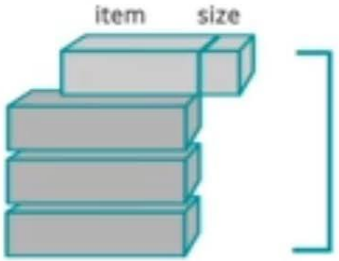
The rest of the Product Backlog emerges to define “what” will fulfill the Product Goal.

A product is a vehicle to deliver value. It has a clear boundary, known stakeholders, well-defined users or customers.

May refer other artifacts like specifications, mockups, architecture models.

Valid product backlog items are: Feature definition, Constraints, Behaviours, Desirements, User actions/stories, Bugs/defects, Use cases, Non-functional requirements.

Product Backlog



Product Backlog Items
(PBIs)

- Features
  - User Stories
  - Use cases
  - Free form text
- Enhancements
- Defect
- Fixes

Product Backlog

#	Backlog Item (User Story)	Story Point
1	As a Teller I want to be able to find clients by last name, so that I can find their profile faster	4
2	As a System Admin I want to be able to configure user setting so that I can control access	2
3	As a System Administrator I want to be able to add new users when required so that...	2
4	As a Data Entry Clerk I want the system to automatically check my spelling so that...	1

Product Backlog

Sprint Backlog

Commitment: Sprint Goal

The Sprint Backlog is composed of the Sprint Goal (why), the set of Product Backlog items selected for the Sprint (what), as well as an actionable plan for delivering the Increment (how).

The Sprint Backlog is a plan by and for the Developers.

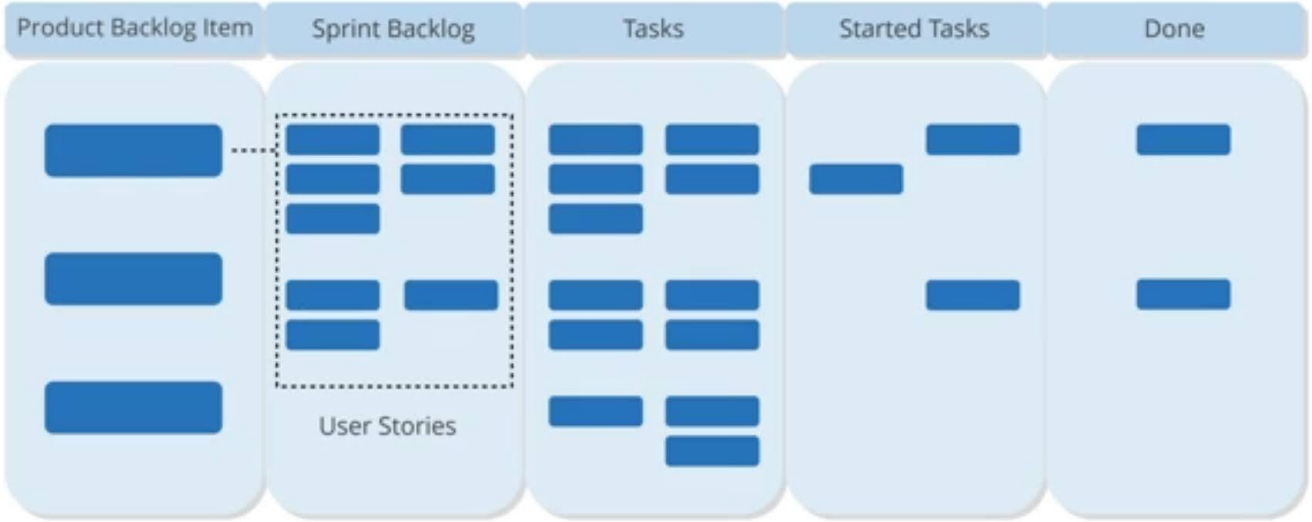
Consequently, the Sprint Backlog is updated throughout the Sprint as more is learned.

The Sprint Goal is created during the Sprint Planning event and then added to the Sprint Backlog.

If the work turns out to be different than they expected, they collaborate with the Product Owner to negotiate the scope of the Sprint Backlog within the Sprint without affecting the Sprint Goal.

Sprint Backlog

Emerges during the Sprint Planning and does not change during the course of Sprint



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Increment

Commitment: Definition of Done

The Definition of Done is a formal description of the state of the Increment when it meets the quality measures required for the product.

The moment a Product Backlog item meets the Definition of Done, an Increment is born.

If a Product Backlog item does not meet the Definition of Done, it cannot be released or even presented at the Sprint Review.

An Increment is a concrete stepping stone toward the Product Goal.

In order to provide value, the Increment must be usable. Multiple Increments may be created within a Sprint.

Definition of 'Done'

Example	
	The "billed" is release ready and available for download.
	Documentation is complete.
	Testing is complete.
	The source code is committed to the server.
	Code has been reviewed and the Product Owner has given approval.

User Stories

User stories are used to describe work items in a non-technical language and from a user's perspective.

As a {type of user}, I want {goal} so that I {receive benefit}.

Example: As a customer, I want to be able to create an account so that I can see my previous purchases.

User Stories

User Story could be a:

Use case - What the user wants to do.

Requirement - Functional requirement, a bug.

User - Fictitious user to explain the requirement

User stories are usually sketched out and prioritized by the product owner, and then the development team determines detailed tasks necessary to complete the story in an upcoming sprint.

The development team is also responsible for estimating the relative effort (story points) required to complete the work of the story.

Once you've created a few user stories, you can start prioritizing them in the backlog.

These are just the starting stories for your project. You will continue to create stories for the project's lifetime. This is because agility involves continuously learning and adapting.

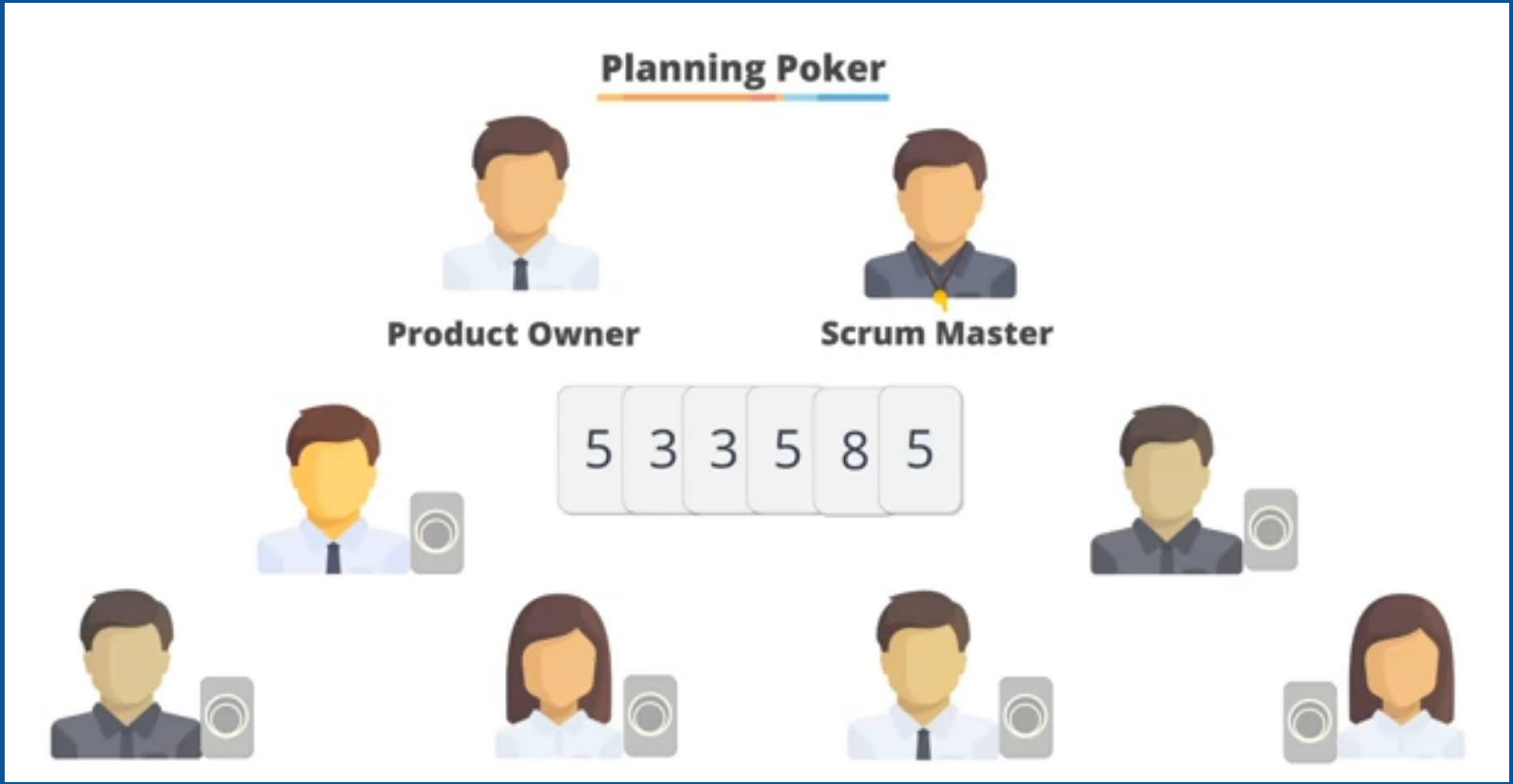
Agile (/User Story) Estimation

Traditional software teams give estimates in a time format: days, weeks, months. Many agile teams, however, have transitioned to story points.

Story points rate the relative effort of work, often in a Fibonacci-like format: 0, 1, 2, 3, 5, 8, 13, 20, 40, 100.

Estimates help you gauge how much work you should add to the next sprint based on the number of team members you have.

After a few sprints, your team will get better at figuring out how much work they can do each sprint, which will help avoid over-committing.


Planning Poker
Product Owner
Scrum Master
5 3 3 5 8 5

Planning Poker
Describes and clarifies
Product Owner
Coaches and facilitates
Scrum Master
Estimates collaboratively
Development Team

Story Point Vs. Ideal Time

Story Points		Ideal Time
• Story points help drive cross-functional behavior. • Story point estimates do not decay. • They are a pure measure of size. • Time required for story points estimation is low.	Vs.	• Ideal time may differ between members of even the same team. • They are easier to explain outside the team; story points are more abstract. • They are easier to estimate, but take longer. • Ideal time can compel companies to confront time wasting activities.

Monitoring Agile Projects

Key KPIs in Monitoring Agile Projects

- Sprint Goal Success Rate
- Escaped Defects
- Burn Down Rate
- Velocity
- Team Output
- Satisfaction
- Team Member Turnover

Information Radiator

Task board

Burn down chart

Burn up chart

Niko-niko calendars

Continuous build health indicator

Burndown Chart

A Burndown Chart shows the actual and estimated amount of work to be done in a sprint. The horizontal x-axis in a Burndown Chart indicates time, while the vertical y-axis typically indicates story points.

Use the Burndown Chart to track the total work remaining for a sprint, and to project the likelihood of achieving the sprint goal. By tracking the remaining work throughout the iteration, a team can manage its progress and respond accordingly.

ANTI-PATTERNS TO WATCH FOR IN BURNDOWN CHARTS

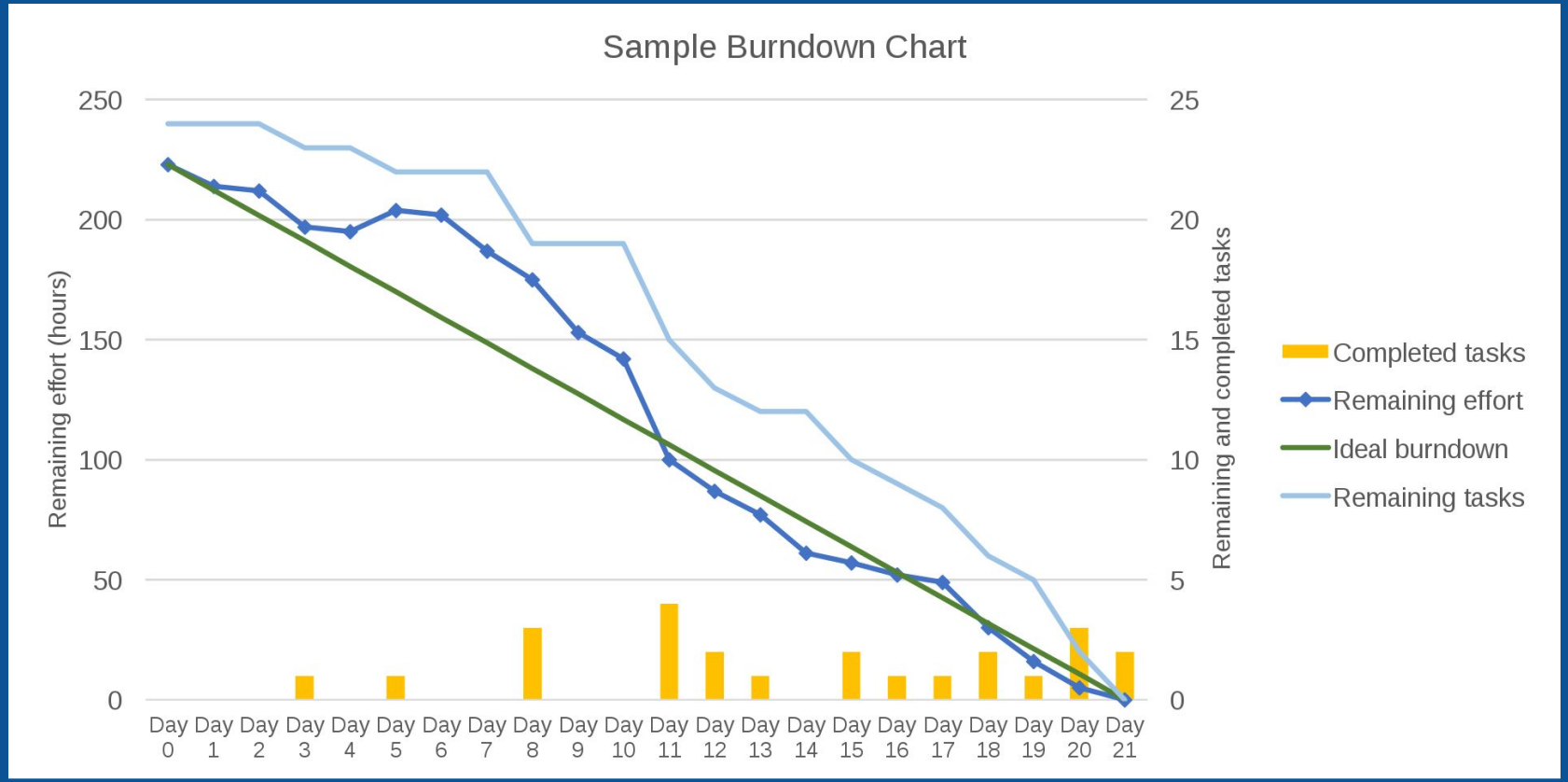
The team finishes early sprint after sprint because they aren't committing to enough work.

The team misses their forecast sprint after sprint because they're committing to too much work.

The burndown line makes steep drops rather than a more gradual burndown because the work hasn't been broken down into granular pieces.

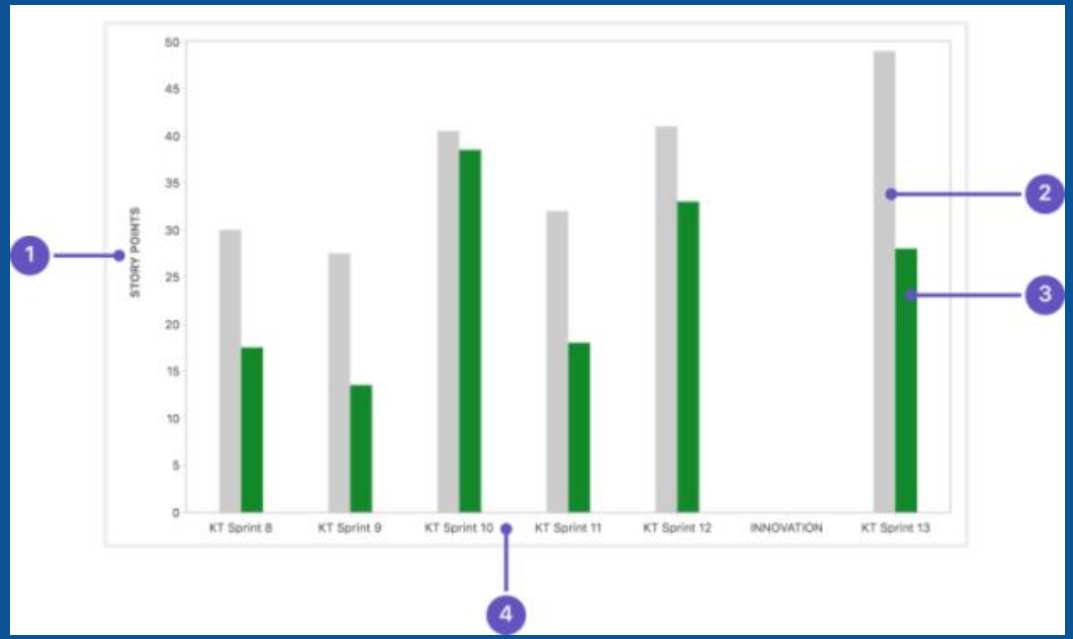
The product owner adds or changes the scope mid-sprint.

Burndown Chart

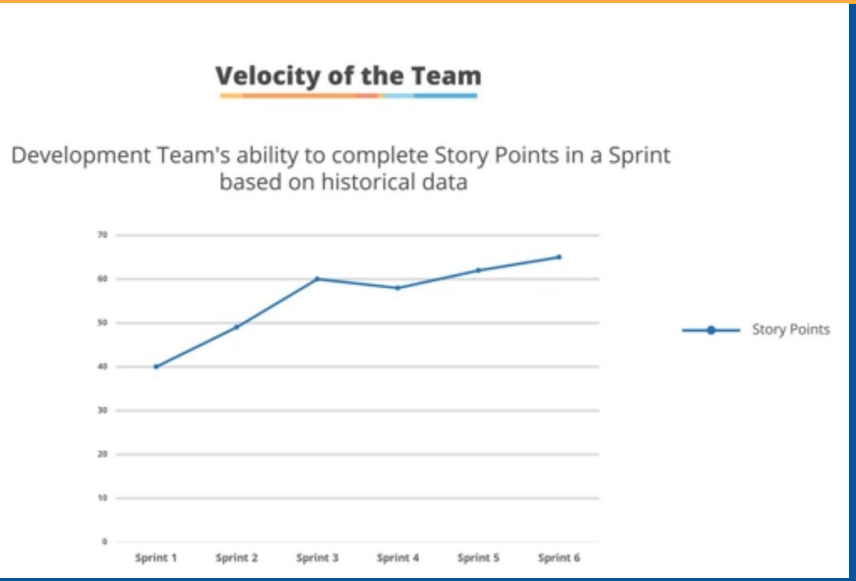


Team Velocity

The Velocity Chart shows the amount of value delivered in each sprint, enabling you to predict the amount of work the team can get done in future sprints. It is useful during your sprint planning meetings, to help you decide how much work you can feasibly commit to.



Team Velocity



Velocity of the Team

Development Team has completed 136 Story Points in the past 12 months

Velocity of the Team = 136 Story Points/12 Months

= 11.33 point per Sprint, rounded off to 12

Different Agile Methodologies

Scrum - Time boxed, Kanban - Event driven, XP (Extreme Programming), Lean, Crystal.

SOS - Scrum Of Scrums.

Agile At Scale:

Scaled Agile framework (SAFe), Large Scale Scrum (LeSS), Disciplined Agile Delivery (DAD), Nexus, Spotify Model, Scrum At Scale.

Why have a Definition of Ready?

- So everyone knows when a User Story is *really* ready to be taken into a Sprint
  - It does not need to be “100% defined” with all acceptance criteria, etc.
  - It does need to be “ready enough” so that the team is confident they can successfully deliver

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Clip slide

Definition of Ready for a User Story

- User Story defined
- User Story Acceptance Criteria defined
- User Story dependencies identified
- User Story sized by Delivery Team
- Scrum Team accepts UE artefacts
- Performance criteria identified, where appropriate
- Person who will accept the User Story is identified
- Team has a good idea what it will mean to Demo the User Story

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Definition of Ready for a Sprint

- The Sprint Backlog is prioritized
- The Spring Backlog contains all defects, User Stories and other work that the team is committing to
  - No hidden work
  - Examples of 'other work' might include lab setup, build environment maintenance, creating a test app
- All team members have calculated their capacity for the Sprint
  - Fulltime on project = X hours per day
- All User Stories meet Definition of Ready